

# **A Cloud Native Multi Modal AI Platform for Explainable Financial Fraud Detection**

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## **Project Proposal Report**

Wijesinghe L P D B – IT22109194

B.Sc. (Hons) Degree in Information Technology Specialized in  
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

| Name               | Student ID | Signature                                                                           |
|--------------------|------------|-------------------------------------------------------------------------------------|
| Wijesinghe L P D B | IT22109194 |  |

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

| Name                | Date | Signature                                                                           |
|---------------------|------|-------------------------------------------------------------------------------------|
| Mrs. Anjalie Gamage |      |  |

## ABSTRACT

Financial fraud in mobile money services has evolved from opportunistic theft to sophisticated algorithmic schemes. Conventional detection systems apply a single global anomaly detection model with uniform thresholds across all transaction types, producing excessive false positives and providing no forensic attribution to explain why a transaction was flagged.

This research introduces a Stratified Variational Autoencoder with Dual-Signal Anomaly Attribution (DSAA) and Unsupervised Fraud Typology Discovery as a behavioral analysis module for the DeepSentinel multi-modal fraud detection platform. The system trains a separate VAE for each major transaction type, learning type-specific normal behavioral patterns. DSAA decomposes both reconstruction error per input feature and KL divergence per latent dimension into interpretable Anomaly Fingerprints that explain which behavioral aspects triggered each alert. These fingerprints are then clustered using DBSCAN to automatically discover distinct fraud typologies without supervision, transforming the system from a per-transaction anomaly detector into a fraud pattern discovery engine.

The system is validated on the PaySim mobile money simulation dataset containing over six million labeled transactions across five transaction types. A seven-configuration ablation study evaluates the contribution of stratification, per-type thresholds, and individual features, including a feature necessity analysis that isolates whether multi-dimensional behavioral modeling adds detection value beyond simple balance verification. Clustering validation uses silhouette score and adjusted Rand index to assess whether discovered typologies correspond to meaningful fraud categories. The system is implemented using TensorFlow/Keras for VAE modeling, scikit-learn for clustering and evaluation, FastAPI for fusion engine integration, and Google Colab Pro for GPU-accelerated training.

**Keywords:** Variational autoencoder, anomaly detection, fraud detection, feature attribution, explainable AI, mobile money, PaySim, unsupervised learning, behavioral profiling, financial forensics, fraud typology discovery, density-based clustering

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## LIST OF ABBREVIATIONS

| Abbreviation | Full Form                                       |
|--------------|-------------------------------------------------|
| AI           | Artificial Intelligence                         |
| API          | Application Programming Interface               |
| ATO          | Account Takeover                                |
| AUC          | Area Under the Curve                            |
| DSAA         | Dual-Signal Anomaly Attribution                 |
| FPR          | False Positive Rate                             |
| GAN          | Generative Adversarial Network                  |
| KL           | Kullback-Leibler (Divergence)                   |
| LIME         | Local Interpretable Model-Agnostic Explanations |
| ML           | Machine Learning                                |
| REST         | Representational State Transfer                 |
| ROC          | Receiver Operating Characteristic               |
| SHAP         | SHapley Additive exPlanations                   |
| SLIIT        | Sri Lanka Institute of Information Technology   |
| SDG          | Sustainable Development Goal                    |
| VAE          | Variational Autoencoder                         |
| WBS          | Work Breakdown Structure                        |
| XAI          | Explainable Artificial Intelligence             |

## **LIST OF APPENDICES**

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# 1. INTRODUCTION

## 1.1 Background and Context

The global financial ecosystem is experiencing a fundamental shift in nature and sophistication of fraudulent activity. Criminal organizations have moved beyond simple unauthorized transactions toward coordinated money laundering schemes that exploit the structural properties of digital payment networks. Techniques such as smurfing where large sums are fragmented into numerous small transactions to avoid reporting thresholds and mule network operations, where chains of compromised accounts are used to layer and consolidate illicit funds, represent modern fraud paradigms that traditional rule-based detection systems were never designed to address.

The scale of this challenge is substantial. The United Nations Office on Drugs and Crime estimates that global money laundering flows represent between two and five percent of world GDP, amounting to eight hundred billion to two trillion US dollars annually. Current detection infrastructure captures less than one percent of these illicit flows, indicating a significant gap between the sophistication of fraud schemes and the capability of existing detection tools.

Within the domain of automated fraud detection, anomaly detection using deep learning has emerged as a promising approach. Autoencoders, which learn compressed representations of normal transaction patterns and flag deviations as potential anomalies, have been widely applied. However, two critical architectural limitations persist across existing implementations.

The first limitation concerns homogeneous modeling. Most autoencoder-based systems train a single global model across all transaction types, forcing the network to learn a compromise representation that inadequately captures the distinct behavioral characteristics of fundamentally different transaction categories. A mobile money payment for consumer goods, a peer-to-peer fund transfer, and a cash withdrawal operation exhibit entirely different normal behavioral distributions in terms of typical amounts, balance changes, temporal patterns, and destination account characteristics. A single model that attempts to represent all these simultaneously inevitably produces elevated false positive rates for transaction types whose normal patterns differ significantly from the global average.

The second limitation concerns explainability. Existing systems produce a single scalar anomaly score that communicates the degree of deviation but provides no forensic attribution regarding which specific aspects of the transaction triggered the alert. This opacity reduces investigator efficiency, erodes trust in automated systems, and creates compliance difficulties under regulatory frameworks that increasingly mandate explainable artificial intelligence.

This research addresses both limitations through a Stratified Variational Autoencoder with Dual Signal Anomaly Attribution, forming the behavioral analysis component of the DeepSentinel multi-modal fraud detection platform.

## 1.2 Literature Review

Anomaly detection using autoencoders has been extensively studied in the context of financial fraud. The foundational work by Kingma and Welling [1] established the Variational Autoencoder framework, which extends standard autoencoders by imposing a structured Gaussian prior on the latent space, enabling both generative modeling and principled anomaly scoring through reconstruction probability and KL divergence measurements. Their architecture provides two independent anomaly signals by construction: reconstruction error measuring input, output fidelity and KL divergence measuring deviation from the learned latent prior. Bowman et al. [4] demonstrated the use of continuous latent spaces for generative modeling, providing theoretical underpinning for the VAE's capacity to model complex behavioral distributions. However, Kingma and Welling's work was theoretical in scope and did not address fraud detection or feature-level forensic attribution, leaving the forensic potential of dual signal decomposition unexplored.

An and Cho [2] subsequently demonstrated that VAE based anomaly detection using reconstruction probability provides a theoretically grounded alternative to threshold-based approaches. Their study confirmed the practical viability of VAE reconstruction error as an anomaly signal and showed superior performance over standard autoencoders on benchmark anomaly datasets. Nevertheless, An and Cho applied a single global model across all data and relied on a single scalar anomaly score, discarding the KL divergence signal entirely and providing no per-feature attribution that would allow investigators to understand which behavioral dimensions triggered an alert.

Autoencoders have been used in the specific field of financial fraud detection to deal with credit card transaction monitoring, insurance claim analysis, and banking fraud identification. In such implementations, the autoencoders are only used to train on legitimate transactions to define a learned representation of normal behavioral patterns. From the reviewed literature, there is a standard limitation in stratification and explainability.

The explainability gap has garnered significant research interest. The latent space interpretability of autoencoders was formally explored for financial crime detection in [5], a 2025 Master's thesis from TU Delft. The research quantified the relationship between latent dimensions and transaction behavior using explicitness, modularity, and compactness. This research is a significant step forward in understanding the latent space of VAEs. However, it is a post-hoc, analytical framework. The framework is used to inform model designers on how to set hyperparameters. However, it is not used to generate attributions for transactions during inference. The Dual Signal Anomaly Attribution framework is a practical, native framework that is used to decompose signals during inference to generate Anomaly Fingerprints for investigators.

A 2025 study combining VAE with multi-head attention mechanisms for credit card fraud detection (Bal-VAE-Attention) [6] demonstrated that incorporating attention layers can improve detection performance on imbalanced datasets. The authors identified that existing methods neglect feature importance mining and that shallow attention structures fail to capture complex data characteristics. Although this work advances the state of the art, its explainability is derived

from attention weights rather than from the VAE's own mathematical structure. Attention weights indicate which inputs the model focuses on, but they do not decompose the VAE reconstruction error and KL divergence into per-component contributions. DSAA, by contrast, requires no additional attention layers: the explanations emerge directly from the VAE's architecture, making it computationally lighter and more theoretically transparent.

A VAE-based fraud detection framework, presented at an IEEE conference in 2025 (VAE-TF) [7], utilizes PCA for dimensionality reduction and SMOTE for class imbalance. The authors employed a ranking-based method where the transactions with the highest errors are selected. Although the effectiveness of the integration of the preprocessing steps is demonstrated by the VAE-TF framework, it solely depends on the errors, providing no attribution at all. This is exactly the problem that the DSAA aims to resolve. While the VAE-TF framework ignores the KL divergence completely, the current work utilizes the two signals independently to obtain the two-level forensic fingerprint. A comprehensive systematic review of 57 research articles on financial fraud detection with deep learning techniques, published by researchers at Georgia Tech in 2025 (Georgia Tech Review) [8], indeed confirmed that the problem of interpretability and explainability of models is one of the main open research challenges in the field. The review indicated that the growth rate of LSTM application is the fastest, while GNNs, GANs, and VAEs are emerging in relational and generative tasks. Most importantly, the review did not find any research articles on the application of native dual-signal decomposition of VAE outputs for forensic attribution on a transaction-by-transaction basis, which indeed supports the fact that the DSAA framework is a new concept not found in the literature.

The PaySim dataset, proposed by Lopez Rojas and authors in [3], offers a research-grade simulated environment for mobile money services based on real aggregate transaction patterns from a multinational mobile money service provider. The PaySim dataset contains more than six million transactions in five types and has labeled fraudulent transactions concentrated in transfer and cash-out types of transactions. The 0.13% fraud rate in the PaySim dataset reflects the realistic class distribution in the system.

Across the VAE-specific literature, no existing work combines stratified per-type modeling with native dual-signal decomposition for real-time forensic attribution the precise gap that this research addresses.

### 1.3 Research Gap

The review of existing literature identifies three specific gaps that this research directly addresses.

Firstly, there is a lack of stratified unsupervised anomaly detection on financial transaction data. While stratified or ensemble methods have been proposed in supervised classification, the use of separate autoencoder models to train on each transaction type to perform unsupervised behavioral profiling has not been adequately studied in the literature on financial fraud detection. The novelty

of the proposed stratified VAE training with adaptive thresholds, supported by empirical threshold analysis and ablation studies, cannot be found in the literature.

Second, the native dual-signal decomposition capability of Variational Autoencoders has not been exploited for per-component forensic attribution in financial fraud detection. While VAEs are used for anomaly detection in prior work [2, 6, 7], the standard practice reduces their output to a single combined anomaly score, discarding the rich per-feature and per-latent-dimension attribution information that the architecture inherently provides. The TU Delft thesis [5] provides an analytical treatment of the interpretability of the latent space, without the generation of fingerprints on a transaction-by-transaction basis at inference time. Bal-VAE-Attention [6] provides attention-based explainability, which is exogenous to the VAE's inherent signal. VAE-TF [7] relies only on the use of the reconstruction error, without the use of the KL term at all. The proposed Dual Signal Anomaly Attribution framework overcomes all three limitations simultaneously, as it is inherent to the VAE architecture, it is performed at inference time, and it decomposes the contributions of the reconstruction error and the KL term into their constituent parts to form a two-level Anomaly Fingerprint specific to each anomalous transaction, which has not been achieved by prior work.

Third, no prior work has demonstrated an explainable behavioral anomaly detection component designed to operate as an orthogonal signal within a multi-modal fraud detection platform. Existing VAE based fraud detection systems function as standalone classifiers, producing a single aggregated risk score that is incompatible with fusion layer architectures that require independently derived, interpretable signals from each component. The behavioral analysis component proposed in this research addresses this gap by exposing a structured Dual Signal Anomaly Attribution output comprising per feature reconstruction attribution and per dimension KL attribution through a REST API contract designed for direct consumption by a fusion engine alongside complementary signals from an edge-enhanced inductive graph learning component (Member 1, GraphSAGE with edge feature integration) and a temporal analysis component. Unlike monolithic detection systems, this architecture ensures that the behavioral signal remains semantically independent, allowing the fusion layer to weight and combine signals without information leakage across detection dimensions.

Fourth, no prior work has utilized native VAE dual-signal decomposition as a feature space for unsupervised fraud typology discovery. Existing systems produce per transaction explanations or anomaly scores but do not aggregate these attribution vectors to discover previously unknown fraud behavior categories. The anomaly fingerprint vectors generated by DSAA decomposition contain rich structural information about why each transaction was flagged. By applying density based clustering on these vectors, the system can automatically group transactions that are anomalous for the same reasons, discovering distinct fraud typologies such as account draining patterns, balance inconsistency patterns, and mule account patterns without any supervision. This capability transforms the system from a per-transaction anomaly detector to a fraud pattern discovery engine, providing investigators with categorical intelligence about the types of fraud activity occurring in the network.

## **2. OBJECTIVES**

### **2.1 Main Objective**

To design and evaluate a Stratified Variational Autoencoder with Dual-Signal Anomaly Attribution that detects behavioral fraud anomalies in mobile money transactions while providing interpretable forensic attribution through per-component anomaly decomposition, and to discover distinct fraud typologies through unsupervised clustering of anomaly fingerprint vectors.

### **2.2 Specific Objectives**

- To engineer behavioral features from PaySim transaction data and stratify transactions by type to enable type-specific behavioral modeling.
- To implement a stratified VAE ensemble where each transaction type receives a specialized Variational Autoencoder trained on type-specific normal behavioral patterns.
- To develop the Dual-Signal Anomaly Attribution (DSAA) framework that decomposes reconstruction error per input feature and KL divergence per latent dimension into interpretable Anomaly Fingerprints.
- To empirically compare per-stratum versus global threshold performance across all transaction types, quantifying false positive rate reduction through stratification.
- To conduct a seven-configuration ablation study comparing global versus stratified models, global versus per-type thresholds, and feature necessity analysis with and without balance consistency features, to empirically validate each system component.
- To apply density-based clustering on dual signal anomaly fingerprint vectors to discover distinct fraud behavior typologies without supervision, validated using silhouette score, adjusted Rand index, and alignment with known PaySim fraud patterns.
- To expose behavioral risk scores, anomaly fingerprints, fraud typology labels, and risk classifications via a REST API for integration with the DeepSentinel multi-modal fusion engine.

## 3. METHODOLOGY

### 3.1 Research Approach

This study follows a design-and-development research approach with an experimental evaluation component. The research begins by identifying the limitations of existing autoencoder-based fraud detection, then proceeds through feature engineering, model architecture design, implementation, and systematic evaluation using ablation methodology. The approach is both technical involving VAE training and anomaly attribution implementation and analytical, involving empirical threshold analysis and comprehensive metric-based evaluation.

### 3.2 Technical Methods and Algorithms

The core analytical method is a Variational Autoencoder architecture applied in a stratified ensemble configuration. Each VAE consists of an encoder network that maps input features to a structured Gaussian latent space parameterized by mean and variance vectors, a reparameterization mechanism that enables gradient-based optimization through stochastic sampling, and a decoder network that reconstructs the input from latent samples. The training objective combines reconstruction loss measuring input-output fidelity with KL divergence loss measuring latent space regularity.

Beta-VAE training with KL annealing will be employed to prevent posterior collapse, where the annealing coefficient increases linearly from zero to one over the initial training epochs. The Free Bits technique will provide an additional safeguard by enforcing a minimum information content per latent dimension.

Following DSAA decomposition, the combined anomaly fingerprint vectors from all flagged transactions will be clustered using DBSCAN (Density-Based Spatial Clustering of Applications with Noise). DBSCAN was selected over K-means because the number of fraud typologies is unknown a priori, and DBSCAN naturally identifies noise points representing unique anomalies that do not belong to any established pattern. The epsilon parameter will be selected using the k-distance plot method, where the distance to each point's k-th nearest neighbor is computed, sorted, and plotted to identify the elbow point. Each discovered cluster will be characterized by its centroid fingerprint vector, identifying the dominant features and latent dimensions that define that fraud typology. Clustering quality will be evaluated using silhouette score, and alignment with known PaySim fraud patterns will be assessed using the adjusted Rand index. Additionally, a feature necessity analysis will be conducted by training and evaluating the stratified VAE ensemble with the balance consistency feature (F3) removed, with F3 alone, and with F3 as the only VAE input, to empirically determine whether the multi-dimensional behavioral approach adds detection value beyond simple balance verification.

Feature engineering will transform raw PaySim columns into eight behavioral features capturing amount magnitude, balance manipulation patterns, mathematical consistency, destination account characteristics, and temporal indicators. Stratification will partition the dataset by transaction type prior to model training.

The Dual Signal Anomaly Attribution framework decomposes the total anomaly score into two interpretable components. Signal 1 decomposes reconstruction error per input feature, revealing which specific transaction characteristics deviate from normal patterns. Signal 2 decomposes KL divergence per latent dimension, revealing which learned behavioral patterns are violated. Together these produce a two level Anomaly Fingerprint for each flagged transaction.

| ID | Feature Name              | Formula                                                              | Fraud Pattern Detected                                      |
|----|---------------------------|----------------------------------------------------------------------|-------------------------------------------------------------|
| F1 | Log Amount                | $\log(\text{amount} + 1)$                                            | Magnitude normalization for scale-invariant comparison      |
| F2 | Amount-to-Balance Ratio   | $\text{amount} / (\text{oldbalanceOrig} + 1)$                        | Account draining: high ratio indicates emptying account     |
| F3 | Balance Consistency Score | 1 if $ \text{oldBal} - \text{amount} - \text{newBal}  < 0.01$ else 0 | Mathematical inconsistency in PaySim fraud transactions     |
| F4 | Balance Change Ratio      | $(\text{newBalOrig} - \text{oldBalOrig}) / (\text{oldBalOrig} + 1)$  | Proportional balance impact assessment                      |
| F5 | Destination Balance Ratio | $\text{newBalDest} / (\text{oldBalDest} + 1)$                        | Mule account detection: empty accounts receiving large sums |
| F6 | Hour of Day               | $(\text{step} \bmod 24) / 24$                                        | Temporal behavioral patterns: late-night transaction risk   |
| F7 | Day of Simulation         | $(\text{step} / 24) / 30$                                            | Periodic behavioral pattern detection                       |
| F8 | Is Large Transaction      | 1 if $\text{amount} > \text{P95 of type}$ else 0                     | Type-relative magnitude outlier detection                   |

Table 1: Feature Engineering Specification

### 3.3 Tools and Platforms

The programming language to be used for the proposed system is Python 3.10, as it is the most widely used language for machine learning research and is compatible with all the required frameworks.

The deep learning framework for VAE model development, training, and inference was TensorFlow 2.x with Keras. TensorFlow was preferred over PyTorch for this project because of its production deployment features using TensorFlow Serving, its superior high-level API using Keras for VAE model prototyping, and its native support for a custom training loop needed to implement KL annealing and the Free Bits technique. Although PyTorch is more flexible for research experimentation, TensorFlow's native ecosystem is important for connecting model

training to API deployment, which is a critical factor in the integration context of the DeepSentinel project.

The libraries NumPy and Pandas will be used for data manipulation and feature engineering. The libraries will be useful for the efficient manipulation of the 6.3 million PaySim data using the required vectorized operations. The libraries Scikit-learn will be useful for the evaluation metrics, optimization of the threshold values using precision-recall curve analysis, and data preprocessing utilities such as normalization and test-train split. The libraries Matplotlib and Seaborn will be useful for the evaluation visualizations, including the ROC curves, precision-recall curves, and Anomaly Fingerprint heatmaps, which are essential for the DSAA output format.

Google Colab Pro with GPU acceleration was selected as the training environment over local hardware due to its cost-effectiveness, zero-configuration setup, and direct integration with Google Drive for experiment artifact storage. This eliminates the capital cost of dedicated GPU infrastructure for a research prototype while providing sufficient computational capacity to train the stratified VAE ensemble on the full PaySim dataset within the project timeline.

FastAPI was chosen as the API framework over Flask for several technical reasons directly relevant to this project. FastAPI provides automatic OpenAPI/Swagger documentation generation, which simplifies the API contract negotiation with Member 4's fusion engine. Its asynchronous request handling supports higher throughput for concurrent inference requests. Its native Pydantic data validation enforces the agreed JSON schema for anomaly fingerprint outputs, reducing integration errors. Flask, while simpler, lacks these production-grade features and requires additional libraries to achieve equivalent functionality. The FastAPI endpoint will expose behavioral risk scores, anomaly fingerprints, and risk classifications as a REST API for consumption by the DeepSentinel fusion engine.

### **3.4 Data Collection and Ethical Considerations**

The study will employ the PaySim Mobile Money Simulator dataset, a research-grade publicly available dataset available on the Kaggle platform, specifically designed to simulate mobile money transactions, including fraud transactions. The dataset has a total of 6,362,620 transactions, consisting of five different transaction types, with a total of 8,213 labeled fraud transactions, mainly occurring during TRANSFER and CASH\_OUT transactions.

The fact that the data is completely artificial, with no personal information about actual individuals, eliminates the need to deal with ethical approvals for the study, as it is completely publicly available for research purposes under the standard terms of the Kaggle platform.

The unsupervised training method eliminates the ethical concerns associated with fraud transactions during the training of the model.



### 3.5 Validation Strategy and Performance Metrics

Model performance will be evaluated using standard classification metrics including AUC-ROC, precision, recall, F1-score, and false positive rate. Threshold optimization will use precision-recall curve analysis to identify operating points that maximize fraud detection while minimizing false alarms.

The seven-configuration ablation study will systematically compare:

- Configuration A: Single global VAE with a single global threshold (baseline)
- Configuration B: Single global VAE with per-type thresholds
- Configuration C: Stratified VAEs with a single global threshold
- Configuration D: Stratified VAEs with per-type thresholds (full proposed system)
- Configuration E: Stratified VAEs with per-type thresholds, excluding the balance consistency feature F3 (feature necessity test)
- Configuration F: Balance consistency feature F3 alone with simple threshold classifier, no VAE (trivial baseline)
- Configuration G: Stratified VAEs with per-type thresholds, trained on F3 as the only input feature (VAE-as-F3-detector test)

This design isolates the contribution of each innovation component. Practical validation will assess the informativeness of anomaly fingerprints by examining whether the attributed features align with known PaySim fraud characteristics such as balance inconsistencies and account draining patterns.

## 4. HIGH-LEVEL SYSTEM ARCHITECTURE

### 4.1 Overview of the Proposed System

The proposed behavioral anomaly detection system operates as one component within the broader DeepSentinel multi-modal platform. It consists of six interconnected processing stages that feature engineering and stratification, VAE model training, anomaly detection and scoring, dual signal attribution, typology discovery, and API based integration with the fusion engine. The architecture is modular allowing each stage to be independently improved or replaced without affecting the overall pipeline.

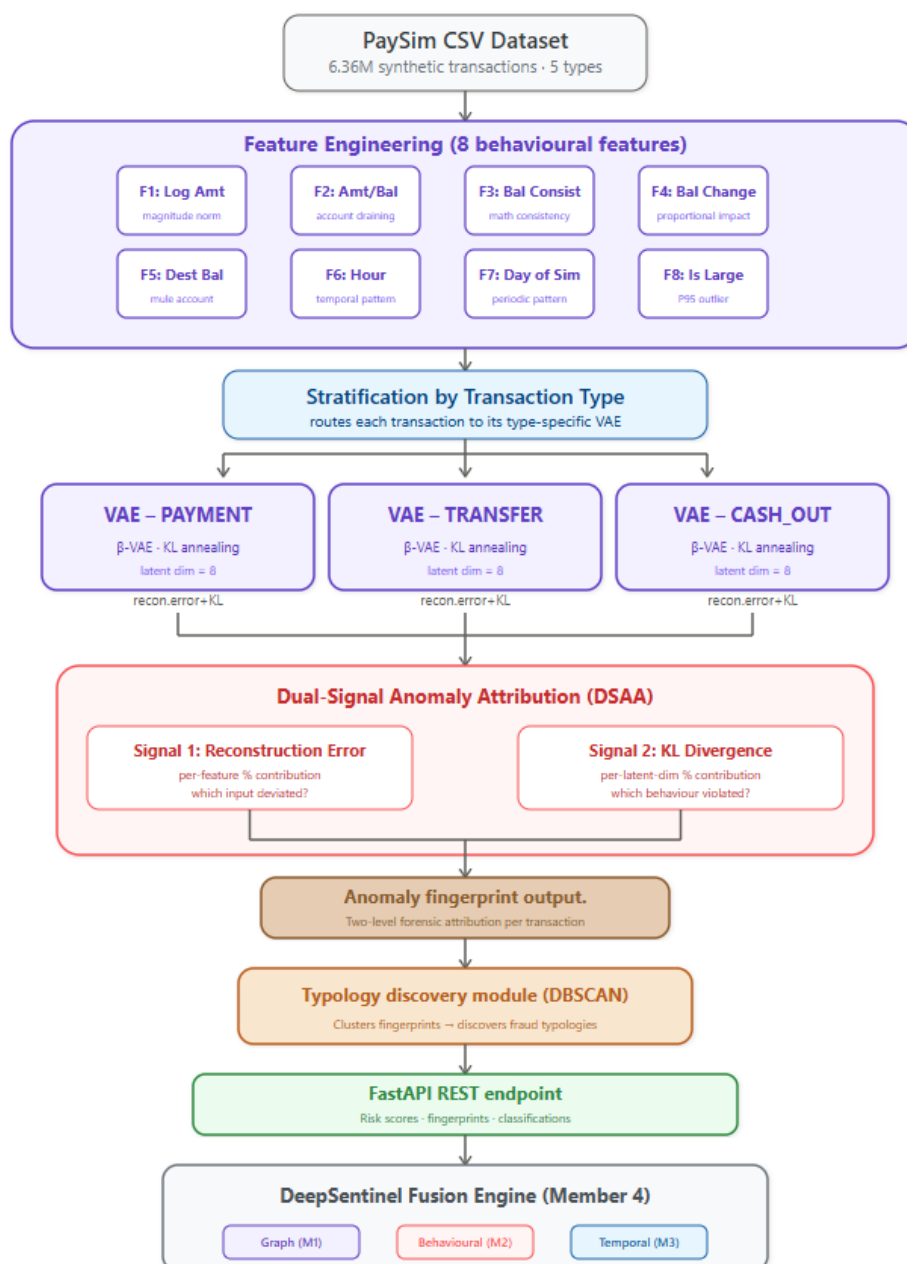


Figure 1: High-Level System Architecture Diagram

## 4.2 System Components

**Feature Engineering Pipeline:** Transforms raw PaySim transaction columns into eight engineered behavioral features and stratifies transactions by type into separate processing streams for PAYMENT, TRANSFER, and CASH\_OUT categories.

**Stratified VAE Ensemble:** Three specialized Variational Autoencoders, each trained exclusively on normal transactions of its assigned type. Each VAE learns type-specific behavioral patterns in a structured Gaussian latent space.

**Anomaly Detection Module:** Computes reconstruction error and KL divergence for incoming transactions using the appropriate type-specific VAE. Applies per-stratum adaptive thresholds to classify transactions as NORMAL, SUSPICIOUS, or CRITICAL.

**DSAA Attribution Engine:** Decomposes reconstruction error per input feature and KL divergence per latent dimension, generating dual Anomaly Fingerprints for each flagged transaction.

**Typology Discovery Module:** Applies DBSCAN clustering on the combined anomaly fingerprint vectors from all flagged transactions to automatically discover distinct fraud behavior typologies. Each cluster is characterized by its centroid fingerprint, identifying the dominant anomalous features and latent dimensions that define that fraud category. Discovered typology labels are appended to the API output for each flagged transaction.

**Integration API:** FastAPI endpoint exposing behavioral risk scores, anomaly fingerprints, and risk classifications for consumption by the DeepSentinel fusion engine.

## 4.3 Data Flow Explanation

| Stage | Stage Name                 | Input                                   | Process                                                    | Output                                                      |
|-------|----------------------------|-----------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| 1     | <b>Data Ingestion</b>      | Raw PaySim CSV file                     | Load 6,362,620 transaction records                         | Raw transaction dataframe                                   |
| 2     | <b>Feature Engineering</b> | Raw transaction columns                 | Compute 8 behavioral features (F1–F8)                      | Enriched feature matrix ( $N \times 8$ )                    |
| 3     | <b>Type Stratification</b> | Enriched feature matrix                 | Route by type: TRANSFER, CASH_OUT, other                   | Separate transaction streams per type                       |
| 4     | <b>VAE Inference</b>       | Type-specific transaction stream        | Pass through trained type-specific VAE encoder and decoder | Reconstruction output + latent encoding ( $\mu, \sigma^2$ ) |
| 5     | <b>Anomaly Scoring</b>     | Reconstruction output + latent encoding | Compute reconstruction error (Signal 1) + KL               | Per-transaction anomaly scores                              |

|   |                             |                                                    |                                                                                                    |                                                                           |
|---|-----------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|   |                             |                                                    | divergence (Signal 2)                                                                              |                                                                           |
| 6 | <b>Threshold Comparison</b> | Anomaly scores                                     | Compare against per-stratum adaptive threshold                                                     | Risk classification: NORMAL / SUSPICIOUS / CRITICAL                       |
| 7 | <b>DSAA Attribution</b>     | Flagged transactions                               | Decompose Signal 1 per feature + Signal 2 per latent dimension                                     | Dual Anomaly Fingerprint per transaction                                  |
| 8 | <b>Typology Discovery</b>   | Anomaly fingerprints from all flagged transactions | Apply DBSCAN clustering on combined 16-dim fingerprint vectors, characterize each cluster centroid | Fraud typology labels and cluster characteristics per flagged transaction |
| 9 | <b>API Output</b>           | Anomaly Fingerprint + risk level                   | Package into JSON response via FastAPI endpoint                                                    | Structured output to DeepSentinel fusion engine                           |

Table 2: Data Flow Explanation

## 4.5 Scalability and Security Considerations

The modular stratified architecture naturally supports horizontal scalability each VAE can be independently deployed and scaled. Adding a new transaction type requires only training an additional VAE without retraining existing models. The API layer implements stateless inference, enabling load balancing across multiple instances. Security considerations include API key authentication for inter-component communication, rate limiting to prevent inference abuse, and model versioning to support controlled rollbacks.

## 4.6 Multi-Modal Integration Context

Within the DeepSentinel platform, this behavioral analysis component provides one of three orthogonal fraud detection signals. Member 1's Edge-Enhanced GraphSAGE component analyzes transaction network topology to detect mule rings and suspicious connection patterns. Member 3's Temporal Convolutional Network analyzes transaction timing patterns to detect algorithmic mimicry and smurfing rhythms. This component analyzes transaction content characteristics to detect behavioral anomalies in amounts, balances, and transaction patterns. Member 4's fusion engine aggregates all three signals into a unified fraud decision.

## 5. USER REQUIREMENTS

### 5.1 Stakeholder Identification

The proposed system serves multiple stakeholder groups within the financial fraud investigation workflow. Financial investigators and compliance analysts are the primary operational users who receive fraud alerts with anomaly fingerprints and must determine appropriate response actions. Risk management officers use aggregated detection analytics to assess institutional exposure and system effectiveness. System integration engineers at Member 4's fusion layer consume the API outputs. Research supervisors and academic evaluators assess the technical contribution and experimental rigor of the proposed approach.

### 5.2 Functional Requirements

The following table defines the functional requirements that the behavioral anomaly detection system must satisfy.

| ID  | Requirement          | Description                                                                                                                                                               |
|-----|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR1 | Data ingestion       | The system shall accept raw PaySim transaction records and compute eight engineered behavioral features per transaction.                                                  |
| FR2 | Type stratification  | The system shall stratify transactions by type and route each to the appropriate specialized VAE model.                                                                   |
| FR3 | Anomaly scoring      | The system shall produce a behavioral anomaly score combining reconstruction error and KL divergence for each transaction.                                                |
| FR4 | Signal 1 attribution | The system shall generate per-feature reconstruction attribution (Signal 1) showing the percentage contribution of each input feature to the total anomaly.               |
| FR5 | Signal 2 attribution | The system shall generate per-dimension KL attribution (Signal 2) showing the percentage contribution of each latent dimension to the total KL divergence.                |
| FR6 | Risk classification  | The system shall classify transactions into risk levels (NORMAL, SUSPICIOUS, CRITICAL) based on per-stratum adaptive thresholds.                                          |
| FR7 | API exposure         | The system shall expose all outputs via a REST API endpoint compatible with the DeepSentinel fusion engine.                                                               |
| FR8 | Ablation support     | The system shall support the four-way ablation study by maintaining all model configurations simultaneously.                                                              |
| FR9 | Typology discovery   | The system shall cluster anomaly fingerprint vectors using DBSCAN to discover distinct fraud behavior typologies and assign a typology label to each flagged transaction. |

Table 3: Functional Requirements

### 5.3 Non-Functional Requirements

The following table defines the non-functional requirements that govern system quality, performance, and integration.

| ID   | Requirement      | Specification                                                                                                                                                             |
|------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR1 | Performance      | The system should process individual transactions within acceptable latency for batch analysis workflows, targeting inference time under 50 milliseconds per transaction. |
| NFR2 | Scalability      | The architecture should support the full PaySim dataset of over six million transactions and be extensible to additional transaction types or feature sets.               |
| NFR3 | Reliability      | The VAE training pipeline should include beta-annealing and free bits safeguards to prevent posterior collapse with at least 98 percent reliability.                      |
| NFR4 | Maintainability  | Model versions, threshold configurations, and feature definitions should be configurable without requiring architectural changes.                                         |
| NFR5 | Interoperability | API output format should conform to the agreed JSON contract with Member 4's fusion engine.                                                                               |

*Table 4: Non-Functional Requirements*

## **6. COMMERCIALIZATION PLAN**

### **6.1 Target Market and Customer Persona**

The primary target market includes mobile money operators, digital payment platforms, commercial banks, and financial regulatory bodies that require advanced fraud detection capabilities with explainable outputs. The main customer persona is the fraud operations manager responsible for detection system performance, false positive reduction, and regulatory compliance with explainability mandates.

### **6.2 Value Proposition**

The proposed system provides explainable behavioral fraud detection that reduces false positive rates through type-aware stratified modeling while delivering two-level forensic attribution that meets regulatory explainability requirements. Unlike black-box anomaly scores, the Anomaly Fingerprints provide investigators with actionable insight that directly reduces mean time to resolve fraud alerts.

### **6.3 Revenue Model**

The behavioral detection component can be commercialized as a module within a larger fraud detection platform, offered through a Software-as-a-Service (SaaS) subscription model. Revenue streams include platform licensing fees, integration consulting, model customization for institution-specific transaction patterns, and periodic model retraining services.

### **6.4 Pricing Strategy**

A tiered pricing model based on transaction volume and feature requirements would accommodate institutions of varying sizes. The basic tier provides core anomaly detection; the standard tier adds dual-signal attribution; the enterprise tier includes custom model training, API integration support, and dedicated retraining schedules.

### **6.5 Competitive Advantage**

The system offers three distinct competitive advantages. The stratified modeling approach eliminates cross-type false positives that plague single-model systems. The native dual-signal attribution provides forensic explainability without the computational overhead and approximation errors of external attribution methods. Empirical validation through a dedicated threshold comparison study quantifies the false positive rate reduction achieved through stratification, providing evidence-based justification that strengthens credibility with technically sophisticated clients.

### **6.6 Intellectual Property Considerations**

Potential intellectual property components include the DSAA attribution framework methodology, the trained stratified VAE model weights, the per-stratum threshold optimization algorithm, and the integrated software implementation.

## **6.7 Cost Estimation**

The estimated cost for developing and evaluating the research prototype is approximately Rs. 33,000, structured across cloud GPU resources, development tools, API infrastructure, storage services, internet connectivity, documentation tools, testing infrastructure, and a contingency reserve. A detailed breakdown is provided in Section 7.



## 7. BUDGET AND JUSTIFICATION

| Description                                             | Amount (Rs.)  | Justification                                                                               |
|---------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------|
| Cloud GPU / Model Training (Google Colab Pro, 6 months) | 8,000         | Required for VAE training, hyperparameter tuning, ablation experiments on 6.3M transactions |
| Cloud Hosting for API Deployment                        | 4,000         | FastAPI endpoint hosting for integration testing with fusion engine                         |
| Development Tools and Libraries                         | 3,000         | PyTorch, scikit-learn, visualization tools, IDE subscriptions                               |
| Database and Storage Services                           | 2,500         | Model checkpoint storage, experiment tracking, result databases                             |
| Internet and Utilities                                  | 3,000         | Development connectivity and power allocation                                               |
| Documentation and Reporting Tools                       | 2,000         | LaTeX, diagram tools, report generation                                                     |
| Testing and Validation Infrastructure                   | 3,500         | End-to-end testing, API load testing, metric validation                                     |
| Human Effort Allocation                                 | 4,000         | Research time for literature review, implementation, evaluation, documentation              |
| Miscellaneous / Contingency                             | 3,000         | Unexpected requirements, additional compute, backup resources                               |
| <b>TOTAL BUDGET</b>                                     | <b>33,000</b> |                                                                                             |

Table 5: Budget And Justification

## 8. WORK BREAKDOWN STRUCTURE (WBS)

| ID  | Task Description                                                                     | Timeline               | Deliverable                                         |
|-----|--------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------|
| T1  | Literature review on VAE anomaly detection, stratified modeling, and financial fraud | March 2026             | Literature review report                            |
| T2  | Feature engineering pipeline development and PaySim data preparation                 | March–April 2026       | Preprocessed stratified dataset                     |
| T3  | Empirical threshold analysis and baseline comparison                                 | April 2026             | Threshold comparison and feature necessity analysis |
| T4  | Global VAE baseline implementation and training                                      | April–May 2026         | Trained baseline model                              |
| T5  | Stratified VAE ensemble implementation and training                                  | May–June 2026          | Three trained type-specific VAEs                    |
| T6  | DSAA framework implementation (Signal 1 + Signal 2)                                  | June–July 2026         | Attribution engine                                  |
| T7  | Per-stratum adaptive threshold optimization                                          | July 2026              | Threshold configuration                             |
| T8  | Four-way ablation study execution and evaluation                                     | July–August 2026       | Ablation results report                             |
| T9  | API development and fusion engine integration                                        | August–September 2026  | Working API endpoint                                |
| T10 | Visualization development and demo preparation                                       | September–October 2026 | Demo-ready prototype                                |
| T11 | Final documentation and presentation preparation                                     | October 2026           | Final research report                               |

Table 6: Work Breakdown Structure

## 9. GANTT CHART

The project timeline spans March to October 2026, aligned with the eleven tasks defined in the Work Breakdown Structure. Semester 1 (March - July) covers literature review, feature engineering, empirical threshold analysis, and VAE model development. Semester 2 (August - October) covers DSAA implementation, evaluation, API integration, visualization, and final reporting.

| ID  | Task                    | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Key                 |
|-----|-------------------------|-----|-----|-----|-----|-----|-----|-----|-----|---------------------|
| T1  | Literature Review       |     |     |     |     |     |     |     |     | <i>Lit. Review</i>  |
| T2  | Feature Engineering     |     |     |     |     |     |     |     |     | <i>Dataset</i>      |
| T3  | Threshold Analysis      |     |     |     |     |     |     |     |     | <i>Threshold</i>    |
| T4  | Global VAE Baseline     |     |     |     |     |     |     |     |     | <i>Baseline</i>     |
| T5  | Stratified VAE Ensemble |     |     |     |     |     |     |     |     | <i>3 VAEs</i>       |
| T6  | DSAA Framework          |     |     |     |     |     |     |     |     | <i>Attribution</i>  |
| T7  | Threshold Optimization  |     |     |     |     |     |     |     |     | <i>Thresholds</i>   |
| T8  | Ablation Study          |     |     |     |     |     |     |     |     | <i>Results</i>      |
| T9  | API Development         |     |     |     |     |     |     |     |     | <i>API Endpoint</i> |
| T10 | Visualizations & Demo   |     |     |     |     |     |     |     |     | <i>Demo Ready</i>   |
| T11 | Final Documentation     |     |     |     |     |     |     |     |     | <i>Final Report</i> |

Figure 2: Grantt Chart

## 10. REFERENCES LIST

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## 11. APPENDICES

### 11.1 VAE Architecture Specification

| Layer              | Type                                                           | Output Shape | Purpose                               |
|--------------------|----------------------------------------------------------------|--------------|---------------------------------------|
| Input              | Input Layer                                                    | (batch, 8)   | 8 engineered features per transaction |
| Encoder Hidden 1   | Dense(32, ReLU)                                                | (batch, 32)  | First compression layer               |
| Encoder Hidden 2   | Dense(16, ReLU)                                                | (batch, 16)  | Second compression layer              |
| Latent Mean        | Dense(8, Linear)                                               | (batch, 8)   | Mean vector of latent Gaussian        |
| Latent Log-Var     | Dense(8, Linear)                                               | (batch, 8)   | Log-variance of latent Gaussian       |
| Reparameterization | $z = \text{mean} + \exp(0.5 * \text{logvar}) * \text{epsilon}$ | (batch, 8)   | Stochastic sampling for gradient flow |
| Decoder Hidden 1   | Dense(16, ReLU)                                                | (batch, 16)  | First reconstruction layer            |
| Decoder Hidden 2   | Dense(32, ReLU)                                                | (batch, 32)  | Second reconstruction layer           |
| Output             | Dense(8, Sigmoid)                                              | (batch, 8)   | Reconstructed 8 features              |

### 11.2 Risk Analysis

| Risk                                   | Impact | Likelihood | Mitigation Strategy                                                                                                                                            |
|----------------------------------------|--------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VAE posterior collapse                 | High   | Low        | Beta-annealing with KL warm-up from 0 to 1 over 10 epochs. Free Bits technique enforces minimum 0.1 KL per dimension. Fallback: use reconstruction error only. |
| PaySim fraud trivially detectable      | Medium | Medium     | Run ablation with and without balance features. Report finding honestly. Demonstrates multi-dimensional vs single-feature detection.                           |
| Stratified model underperforms global  | Medium | Low        | Ablation study includes global baseline. Negative result is valid research finding with analysis. Global model serves as fallback.                             |
| Class imbalance affects evaluation     | Medium | High       | Train unsupervised on normal data only. Use precision-recall curves instead of accuracy. Report AUC as primary threshold-independent metric.                   |
| Integration failure with fusion engine | Low    | Low        | Agree JSON API contract early. Test with curl before integration. Fallback: expose as Python function import.                                                  |
| Insufficient computational resources   | Low    | Medium     | Use Google Colab Pro. Train on stratified samples if full dataset exceeds memory. Batch processing.                                                            |

|                                        |        |        |                                                                                                                                                                                                                                                                           |
|----------------------------------------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fingerprint clusters not interpretable | Medium | Medium | Validate discovered clusters against known PaySim fraud subtypes (balance drain, account emptying). If clusters are homogeneous, report as a valid finding. Use t-SNE visualization to qualitatively assess cluster separation. Hierarchical clustering as backup method. |
|----------------------------------------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 11.3 Ethical Considerations

This research uses the PaySim dataset, which is entirely synthetic and contains no personally identifiable information. No real financial data, customer records, or banking credentials are involved at any stage. The dataset is publicly available on Kaggle under standard research usage terms. The proposed system is designed as a decision support tool that assists human investigators rather than making autonomous decisions, ensuring that final fraud determinations remain under human oversight and accountability. The research does not involve human participants and therefore does not require formal ethical clearance from the institutional review board.